



P0b

Structure Plus Scale Is the Intelligence: Everything Has Scaling Laws

结构加规模万物皆可 Scaling Laws

TwinsEarth · 智能体宇宙论文系列

Structure Plus Scale Is the Intelligence: Everything Has Scaling Laws

Working Title (EN): Structure Plus Scale Is the Intelligence: A Three-Section Scaling Law Across Substrates

Flagship companion paper P0b • Sequel to P0 "Spatial Structure Is the Intelligence" • UDOS Reasoning Engine v7.5.0 • 2026-09-20

In one sentence: P0 argues that "structure determines intelligence," and this paper adds the other half — "scale." Structure determines what the growth curve looks like, and scale determines which section of the curve you stand on; **adding scale on a wrong structure does not make you stronger — it makes you weaker.**

Evidence scope (stated only once in the whole text): the figures produced by the UDOS engine itself all come from `reports7/*.json`, run on an ordinary CPU with fixed random seeds and reproducible. They fall into three grades: `verified` (guarded by automated tests), `cpu-proto` (trends seen on a CPU prototype, whose constants have not been calibrated across multiple seeds), and `unverified` (external reports or visions, used only as motivation). Every single-seed pilot is explicitly labeled a "pilot," and ≥ 30 seeds are required for recomputation before submission. External literature was verified item by item on 2026-09-20.

Abstract

Over the past decade, the success of artificial intelligence can be summed up in almost one word: go big. Larger models, more data, and ample compute — and loss falls along a beautiful power-law curve. But this "bigger is better" narrative secretly assumes one thing: **the structure of the network is fixed** — everyone uses the Transformer, stacks layers in the established way, and only adds scale along this fixed track.

This paper puts the frozen "structure" back in and proposes a proposition that can be killed by experiment. We write capability as a function of two variables: structure \$\$\$ (how things are connected and partitioned) and scale (parameters/nodes \$N\$, data \$D\$, compute \$C\$). Across seven domains — biological metabolism, cities, brains, chip interconnects, and large models — the growth of capability with scale all exhibits the same **three-section curve**: first a steady power-law growth, then a structural phase transition at some critical point, and finally saturation or even a downward turn. **The shape of the curve is determined by structure, and which section of the curve the system stands on is determined by scale.**

On our own multi-agent engine, we saw the prototype of this curve through seven controlled comparisons: with the same one million nodes, the center of a star topology must simultaneously process about 20.97 million messages with a critical path as long as 8.39 million rounds, while in a hierarchical topology each node faces at most 9 messages and finishes in 14 rounds — this is not "a bit more efficiency," but the growth exponent changing from 1 to 0; increasing concurrent agents from 1 to 2 raises throughput by 1.59 $\times$, but continuing to 8 is slower than 1; equal-weight insertion of an unqualified member into the team makes error surge from 0.0092 to 0.2131, while simply increasing the number of qualified members barely moves the error.

This paper does not claim to have found a theory of everything. The harshest objections — "the emergence turning point may be an illusion created by the measurement method" and "a single-seed pilot is not sufficient evidence" — are listed item by item in Chapter 7, along with what results would falsify this paper.

Keywords: scaling law; structure and scale; three-section curve; phase-transition turning point; saturation and negative return; falsifiable prediction; UDOS; multi-agent

1 Starting from Three Simple Phenomena

1.1 Why an elephant is not a thousand mice

An elephant's body mass is about a hundred thousand times that of a mouse, but its metabolic rate is not a hundred thousand times — it is roughly the three-quarter power of a hundred thousand times; when body mass doubles, the energy needed to sustain life increases by only about 75%. This is the pattern the biologist Kleiber discovered in 1932, later called Kleiber's law.

A larger body is not free. Blood vessels must branch again and again to deliver nutrients to every cell; the longer the network, the greater the "structural discount" of transport. **The returns from scale growth are inherently discounted, and the discount rate is determined by the structure of the transport network.**

1.2 Larger cities are more economical, and also more profitable

In 2007 the physicist Bettencourt and colleagues analyzed a large amount of urban data and found a symmetric phenomenon: when urban population doubles, infrastructure such as roads, cables, and gas stations increases by only about 80% (sublinear, economical), while innovation outputs such as patents, wages, and GDP increase by about 120% (superlinear, profitable).

On the same "population scale" axis, different kinds of "structural quantities" give oppositely directed exponents. This shows that scale itself has no fixed good or bad; **good or bad depends on what you are counting and how those things are organized.**

1.3 Two agents are the fastest, eight are the slowest

On our own engine, an even more counter-intuitive phenomenon is directly reproducible. Let concurrent agents process a batch of mutually independent tasks:

- With 1 agent, 153 tasks per second;
- With 2, a peak of 244 ($1.59\times$ speedup);
- With 4, it falls back to 158;
- With 8, only 106 remain — **slower than 1.**

But as soon as the task type is changed from "competing for CPU" to "waiting for external input" (e.g., waiting on the network or disk), increasing from 1 to 8 makes throughput grow almost perfectly linearly to nearly $8\times$.

The scale axis is identical, and the only thing changed is "whether the task structure and the hardware structure match." When they match, adding scale is nearly free; when mismatched, adding scale is making a mess of the system.

These three phenomena — metabolism, cities, and agents — together point to the central thesis of this paper:

Scale is never a good thing, nor a bad thing. The returns of scale are a function of structure.

2 Making It Clear: Two Variables, One Curve

2.1 What structure is, what scale is

Following the definition of companion paper P0, **structure** includes three things:

- **Connection topology**: who connects to whom and in which direction information flows (corresponding to star/hierarchical/mesh in multi-agent systems);
- **Spatial organization**: how connections are geometrically arranged (distance, viewpoint, fan-in/fan-out);
- **Module boundaries**: how the system is layered, encapsulated, and interfaced between modules.

Scale is decomposed into three measurable axes:

- **\$N\$ (nodes/parameters)**: how many agents, how many parameters, how many neurons;
- **\$D\$ (data/experience)**: how many trajectories and tokens have been seen;
- **\$C\$ (compute/energy)**: how many FLOPs, how much power, how much wall-clock time.

Thus capability can be written as a binary function:

$$C \approx F(\text{structure } S; N, D, C)$$

This formula can be understood without a mathematical background: **capability depends simultaneously on "how it is organized" and "how large it is organized."**

2.2 The three claims of this paper

Claim one: structure determines the shape of the curve. On log-log coordinates, how capability grows with scale, where it bends, and where it peaks are all prescribed by structure. Changing the connection mode does not raise the curve a little; it replaces it with a different curve.

Claim two: scale determines which section of the curve you stand on. Once structure is fixed, adding scale only moves along the established curve; it cannot change the curve's shape.

Claim three: when structure is mismatched, the marginal return of scale is negative. When the mode of organization does not match the task or resources, continuing to add people, parameters, or machines makes capability fall rather than rise. Section 1.3's "8 agents are slower than 1" is a direct observation of this claim.

As an analogy: **structure is grammar, scale is length.** With wrong grammar, the longer the writing the more incoherent it becomes; with right grammar, length begins to produce value.

2.3 The three-section curve: turning "everything has" into a falsifiable proposition

If "everything has a scaling law" means only that "any data can be fitted to a power law," it is an unfalsifiable empty phrase. We tighten it into a far more concrete proposition — the **three-section curve**:

Figure 4: Three-section master curve, schematic

Figure 4: A three-section schematic of capability versus scale (a schematic, not measured data). Section 1 is steady power-law growth; Section 2 shows a turning point at the structural phase transition; Section 3 saturates or turns downward. A good structure pushes the turning point further right; a wrong structure makes it arrive early.

- **Section 1 • Local power-law region:** structure is correct and scale is still small; capability grows steadily with scale, approximately a straight line on log-log coordinates;
- **Section 2 • Structural phase-transition turning point:** scale reaches the threshold where the old structure cannot hold, and the system must change its mode of organization (from one center to multi-layer division of labor, from a single expert to routed experts); the curve bends here;
- **Section 3 • Saturation or negative-return region:** after hitting the communication, data, or energy wall, growth flattens or even turns downward.

This proposition can be killed: if, in a controlled experiment with structure strictly fixed, capability keeps growing linearly along the same slope across more than tenfold scale without peaking or bending, the three-section curve is wrong.

3 Nature Has Walked This Path for Three Billion Years

Before discussing our own experiments, let us see whether different domains repeatedly exhibit the same curve. Table 1 is the overview, followed by the most persuasive layers.

Table 1 Overview of scaling phenomena across seven domains (exponents are empirical ranges, not universal constants)

Domain	Structural unit	Scale axis	Observed pattern	Turning point/wall
Chemical bonds	Bond type and crystal lattice	Atoms/unit cells	Properties jump with bond type, not smoothly with atom count	Bond-type switch is a phase transition
Proteins	Sequence → folding	Residues	Small proteins have one domain; large proteins are modular with multiple domains	Heat denaturation, function falls off a cliff to zero
Life metabolism	Fractal transport vessel network	Body mass	Metabolic rate $\propto$ body mass to the 3/4 power (Kleiber's law)	Surface-area/volume wall of cell size
Chips/compute	Interconnect topology	Cores/chip count	Amdahl's law caps it; USL predicts adding machines makes it slower	Communication wall, power wall
Battery cells	Series-parallel and battery management	Cell count	Voltage with series, capacity with parallel are approximately linear	Thermal-runaway threshold (lacks rigorous measurement, left blank)
Brain	Connectome and cortex	Neuron count	The human brain has about 86 billion neurons, scaling linearly by primate rules	White-matter wiring cost, cortical folding
Large models	Attention/routing structure	Parameters/data/compute	Loss falls as a power law with parameters (exponent about 0.076)	"Emergence" turning point (may be a measurement illusion)

3.1 Metabolism: the 3/4 discount rate is computed from vessel structure

The deepest point of Kleiber's law is not the figure itself but its origin. In 1997, West, Brown, and Enquist gave a structural explanation in *Science*: blood vessels are a fractal tree branching from the aorta all the way to capillaries; to keep terminal capillary size independent of body size (because cell size is universal) while minimizing transport energy, the geometry of this tree must satisfy specific self-similar relationships — **the 3/4 power was not assumed by anyone; it is derived from the geometric constraints of the transport network.**

Replace it with a non-branching straight tube and the exponent is no longer 3/4. This is precisely the natural template for claim one: **the scaling exponent is a product of structure, not an accident of scale.**

It must be honestly stated that 3/4 is an empirical approximation rather than an iron law; some scholars point out that it is not statistically significantly better than the naive 2/3, and the exponent drifts between 2/3 and 1 across species and body temperatures. We cite it not to carry 3/4 as a universal constant into AI, but to use it to prove that "deriving the growth exponent from structural geometry" is possible.

3.2 Proteins: denaturation is a phase transition with no intermediate state

A protein's function depends entirely on its three-dimensional folded shape, and folding is a cooperative process: under slow heating nothing happens, but at some narrow temperature interval the hydrogen bonds maintaining the shape collectively break, and the protein jumps in an instant from the native

state to the denatured state, with catalytic function falling off a cliff to zero — there is no "half-inactivated" intermediate state.

This is the cleanest natural sample of the "structural phase-transition turning point" in the three sections: **structure is effective on one side of the turning point and fails on the other**. Moreover, small proteins fold as one whole block, while large proteins are split into multiple independently folded domains linked together — larger scale forces the system to phase-transition from "one whole block" to "modular assembly," the same principle as replacing a single feed-forward network with multiple routed experts (MoE) in large models.

3.3 Brains: long-distance wiring grows faster than computational units

"The human brain has a hundred billion neurons" is a rough estimate that has circulated for half a century. The neuroscientist Herculano-Houzel used an isotropic fractionator to recount, giving about 86 billion neurons, and found that the human brain is not some extraordinary organ but a **primate brain linearly scaled up by standard primate rules**.

More interesting is the allometric growth of white matter versus gray matter: as the brain enlarges, the white matter responsible for long-distance connections grows faster than the gray matter responsible for computation. Wiring cost hits the wall first, and the brain's response is to fold the cortex into wrinkles, shortening connections within a limited skull. **When long-distance communication becomes expensive with scale, the system adds layers and localizes communication** — the same structural move as chips shifting from shared buses to fat-tree hierarchical interconnect, and our replacing a star center with hierarchical topology in the engine.

3.4 Chips: "adding machines makes it slower" has a formula

Engineering has long had mature theories of scale returns turning negative.

- **Amdahl's law (Amdahl, 1967)**: as long as a small fraction of a program cannot be parallelized, no matter how many cores are added the speedup is sealed by that fraction — it explains "peaking."
- **The Universal Scalability Law USL (Gunther, 1993)** goes further, predicting the "downward turn":

$$S_p = \frac{p}{1 + \sigma(p-1) + \kappa p(p-1)}$$

Where p is the number of parallel workers, σ is the cost of serialization and resource contention, and κ is the cost of keeping state consistent among multiple workers, the latter growing **quadratically** with the number of people. When $\kappa > 0$, throughput must monotonically decline after rising to a peak — **the more people, the slower the team**.

Chip history also has an energy wall. The 1974 Dennard scaling once promised "smaller transistors, faster speed, constant power density," driving clock frequency from megahertz to gigahertz; but around 2005–2007, leakage and threshold voltage could no longer shrink proportionally, power density began to spiral out of control, the "free lunch" ended, and the industry was forced to abandon single-core frequency increases for multi-core parallelism (Intel canceling the Tejas project in 2004 was the landmark event). **This itself is a structural phase transition: not that transistors became smaller and slower, but that the path of "continuing single-core frequency stacking" physically hit the wall and structure had to change.**

3.5 Large models: AI is the most thoroughly studied domain of scaling laws

Three works form the backbone on the AI side:

- **Kaplan et al. (2020)**: loss falls approximately as a power law with parameters, data, and compute, with an exponent on parameters of about 0.076 — direct evidence of the Section 1 power-law region;
- **Chinchilla (Hoffmann et al., 2022)**: under fixed compute, parameters and data should grow in the same proportion, and stacking parameters without feeding data is wasteful — this is in fact "mix-ratio structure determines efficiency";
- **Switch Transformer (Fedus et al., 2021)**: replace the feed-forward layer with a set of experts, activating only one per forward pass, reaching about 1.6 trillion total parameters with very small per-pass compute. It proves that **routing structure itself is a new scale axis**: not by growing larger, but by changing "which parameters participate in computation."

DeepSeek's NSA sparse attention (2025) is a more recent case of the same kind: without adding parameters, only changing the connection structure of attention (three branches — compression, selection, sliding window — plus gating), it reduces the decoding latency of 64k long context to about one-eleventh. **What truly changes the scale curve is often a structural switch, not simply piling on materials.**

As for "emergence" — certain capabilities being zero on small models and suddenly appearing at some scale — we must immediately introduce the harshest counter-argument: Schaeffer et al. (2023) prove that many such turning points are **illusions selected by the measurement method**: replacing a non-continuous metric such as "whether 50% accuracy is crossed" with a continuous metric, the capabilities of the same batch of models actually grow smoothly and "emergence" disappears. This criticism determines the restrained attitude of this paper: a turning point can only be called a "candidate turning point" until it is reproduced on deterministic metrics across multiple domains.

3.6 An honest blank

Of the seven layers, "battery cells" is the thinnest in evidence. We only have circuit common sense such as voltage growing linearly with the number of series cells and capacity growing linearly with the number of parallel cells, plus the engineering experience that "series amplifies individual inconsistency and requires a battery management system for balancing"; we do not have a peer-reviewed rigorous scaling exponent between battery-pack scale and performance.

Under the discipline of this paper, **when evidence is insufficient, leave it blank and mark it [CITATION NEEDED] , rather than hard-coding a pretty figure to complete the "seven layers."** This blank itself is a qualification of the words "everything has": it is a program to be tested, not a fact list already filled in.

4 What We Saw on Our Own Engine

This chapter is the original evidence of this paper. All seven experiments follow the same discipline: **change only one of structure or scale at a time, freezing everything else.** All figures are reproducible on a CPU, with sources in the data-provenance table at the end.

4.1 Evidence one: change the connection mode and the growth exponent changes from 1 to 0

Let a star topology (all agents report to one center) and a hierarchical topology (tree-shaped division of labor) handle the same coordination task, changing only the topology, extrapolating from 10 nodes all the way to one million:

Table 2 Star vs hierarchical: changing only connection structure (first two rows measured, last four closed-form extrapolation, grade cpu-proto)

Target nodes	Actually generated nodes	Messages the star center faces	Hierarchical max fan-in	Star critical path (rounds)	Hierarchical critical path (rounds)
10	73	640	9	256	4
100	585	5,120	9	2,048	6
1,000	4,681	40,960	9	16,384	8
10,000	37,449	327,680	9	131,072	10
100,000	299,593	2,621,440	9	1,048,576	12
1,000,000	2,396,745	20,971,520	9	8,388,608	14

Figure 2: UDOS fan-in/rounds log-log

Figure 2: The horizontal axis is node count (log scale). The star topology's center load explodes linearly with scale (exponent 1), the hierarchical topology's per-node load is constantly 9 (exponent 0), and the critical path grows only slowly with the logarithm.

How to read this table: at a million nodes the star center must simultaneously process about 20.97 million messages and walk 8.39 million rounds — this is not "a bit more expensive," it is physically impossible; the hierarchical topology's nodes forever face only 9 neighbors and finish in 14 rounds. The difference between the two is not an efficiency constant but a growth exponent: one explodes linearly with scale, one is constantly constant. **This is the most direct reading of "structure determines the curve shape."**

4.2 Evidence two: on a mismatched structure, adding people makes it progressively slower

The throughput hump of Section 1.3 is worth fixing with a table. The same batch of agents, the same dual-core machine, changing only the task structure:

Table 3 Throughput versus concurrency (single-seed pilot, grade cpu-proto)

Concurrent agents	CPU-intensive (tasks/s)	Speedup	I/O-waiting (tasks/s)	Speedup
1	153.208	1.000	19.94	1.00
2	244.172	1.594	39.88	2.00
4	158.386	1.034	79.69	4.00
8	106.485	0.695	159.26	7.99

Figure 3: Throughput hump

Figure 3: The CPU-intensive task peaks at 2 agents and then turns, with 8 slower than 1; the I/O-waiting task is nearly perfectly linear. The scale axis is the same, and the curve shape is entirely determined by the structural match between task and hardware.

This is precisely the shape of the USL formula (Eq. 2) under two different κ values: when CPU-intensive, agents beyond the number of physical cores contend for compute resources and repeatedly synchronize shared state, with the consistency cost surging quadratically in the number of people and the curve peaking then falling; when I/O-waiting, agents spend their time waiting without contending, and both σ and κ are near zero, making the curve approximately a straight line.

It must be emphasized that this is a **single-seed, four discrete-level** pilot. "Peak at 2" may partly be an apparent turning point caused by the coarse 2→4 level (exactly the measurement illusion Schaeffer warns of), and whether "8 slower than 1" is stable requires denser levels and ≥ 30 -seed recomputation. We list it as candidate evidence, not a settled conclusion.

4.3 Evidence three: a 23× structural difference, only a 0.5% quantity difference

Integrating the judgments of multiple agents, how does quality change with team size? We fit $mse(k)=a+b/k$, obtaining a saturation floor $a=0.009252$, $b=-4.233 \times 10^{-5}$ ($k \leq 2$ is verified, larger k is cpu-proto); the error of the single best member is 0.009196, almost coincident with the saturation floor — the ensemble already peaks near $k=1$.

Figure 5: Quality saturation and model-size convergence

Figure 5: Left, the weighted-ensemble quality of qualified members already hugs the saturation floor near $k=1$, with continued additions giving almost no improvement; wrongly equal-weight inclusion of one unqualified member makes the error jump immediately to 0.2131. Right, the model-width scan selects 967,796 parameters by the "stop when good enough" rule.

The two figures placed together make the conclusion very plain:

- Expanding the team from 1 qualified member to 17 moves the error only about 0.5% around 0.0092 — **the marginal return of quantity is nearly zero;**
- But as soon as one unqualified member is inserted **with equal weight** into the decision, the error jumps immediately from 0.0092 to 0.2131 — **the influence of structure (how to weight, who gets to speak) is about 23× that of quantity.**

When the system already stands in Section 3 (the saturation region), "adding people" is the most ineffective action; "changing structure" is the one. This also explains a common misconception in the multi-agent field: a "legion" piled together by headcount is often inferior to one group with clear division of labor.

4.4 Evidence four: with the same data budget, changing the collection method yields 37% more effective data

The scale axis is not only "how much data is used," but also "how data is collected." Under an identical collection budget (48 collection rounds), compare two collection structures: passive volume piling versus active coverage-aware blind-spot scanning:

Table 4 Active versus passive collection (same budget, grade cpu-proto)

Collection structure	Data yield	Effective state units covered	Share of ideal coverage	Redundancy rate	Experience density (units/min)
Passive collection	0.667	157	0.695	0.599	7.80
Active blind-spot scanning	0.854	215	0.951	0.692	11.38

Without adding a cent to the budget, changing only the structure of "what to collect" raises yield from 0.667 to 0.854 and the effective state units collected per minute from 7.80 to 11.38. **The effective slope of the data scale axis is determined by collection structure.** If the 10-millionth hour is merely a low-quality repetition of the first 1 million hours, what it brings is not growth but a storage bill.

The gain from active scanning must rule out a trivial explanation — "it just greedily picked easy samples." A controlled experiment restricting the Agent to greedy selection within the same biased data pool reaches only a yield of 0.7917 and coverage share 0.889; while active scanning that can schedule new collection and truly fill blind spots reaches 0.854 / 0.951. **The gain comes mainly from "collecting where one has not been," not from "selecting within old data."**

4.5 Evidence five: models are not the bigger the better, but "stop when good enough"

Scanning the hidden-layer width of the world model (64/128/256), performance improves monotonically and parameters grow from 95,000 to 968,000:

Table 5 Model-size scan (grade verified)

Hidden width	Parameters	Validation metric	Blind-test error
64	95,348	0.01190	0.01527
128	271,476	0.00950	0.01172
256	967,796 (selected)	0.00821	0.01037

The selection rule is "take the **smallest** model whose metric is within a 3% tolerance of the optimal value." In this scan performance was still improving, so the largest level was selected; but the rule itself embodies the Chinchilla spirit — **scale should match the task, rather than mindlessly taking the largest.** Whether saturation or even degradation appears at wider widths (512/1024) is not supported by this data and is marked [RESULT NEEDED] .

4.6 Evidence six: governance also has a scale tax, and structure can bargain it down

As scale grows, cost lies not only in compute but in coordination and trust. Three small experiments:

- **Byzantine fault tolerance:** to tolerate f malicious nodes, the committee size must satisfy $n \geq 3f+1$ with a $2f+1$ quorum — security carries a scale tax; under $n=7, f=2$, all 24 tasks were adjudicated correctly;
- **Internal market accounting:** budget, payment, and balance are strictly conserved ($240 = 240 = 240$); conservation is a cross-scale invariant, and mismatched accounts indicate something wrong with system state;
- **Environmental-medium collaboration (stigmergy):** with the same 30 tasks and 6 agents, collaborating by leaving traces on one shared blackboard needs only 60 messages, while peer-to-peer negotiation one by one needs 240 — **changing the communication structure from "pairwise dialogue" to "environmental note-taking" reduces communication volume to a quarter.**

4.7 Evidence seven: spatial structure also saturates

Predicting a scene from an unfamiliar viewpoint, using only 1 training viewpoint, the contour overlap IoU is 0.7147, spatial coverage 0.7456, precision 0.9472, depth error MAE 0.3186 (voxel resolution 0.1 m, 6 held-out viewpoints); using 8 pose-tagged viewpoints raises IoU to 0.8897 and coverage to 0.9957. But split by viewpoint, **the first two viewpoints contribute the vast majority of the gain, and later viewpoints' marginal contribution rapidly shrinks** — once the viewpoint structure is correct, a few viewpoints approach saturation, and continuing to pile on viewpoints yields little.

4.8 The shape assembled from seven pieces of evidence

Table 6 Overview of seven pieces of evidence

#	Experiment	Changed structure	Scale axis	Key figures	Grade
1	Star vs hierarchical	Topology	Node count	Center load 20.97 million → 9; rounds 8.39 million → 14	cpu-proto
2	Throughput hump	Task structure	Concurrency	2 peak, 8 slower than 1	cpu-proto
3	Quality saturation	Weighted/equal	Member count	23× structural difference, 0.5% quantity difference	verified/cpu-proto
4	Data flywheel	Collection structure	Data budget	Yield 0.667→0.854	cpu-proto
5	Model selection	Width	Parameters	"Stop when good enough" selects 968,000 parameters	verified
6	Governance scale tax	Communication/trust structure	Node count	Messages 240→60; $\$n\geq 3f+1\$$	cpu-proto
7	Multi-view	Spatial structure	Viewpoint count	IoU 0.7147→0.8897, diminishing gains	cpu-proto

Putting them back into the three sections: the hierarchical side of evidence 4 and evidence 1 falls in Section 1 (correct structure, small scale, steady growth); the star→hierarchical switch of evidence 1 and the negotiation→blackboard switch of evidence 6 fall in Section 2 (structural phase transition pushes the turning point open); the hump of evidence 2, the saturation of evidence 3, and the marginal decline of evidence 7 fall in Section 3. **The seven fragments assemble the prototype of the three-section curve on silicon agents — only a prototype.**

5 How to Prove This Paper Wrong

The value of a scientific proposition lies in what can kill it. From the original five predictions we retain the three hardest, each with its criterion written out.

Prediction one (reproducible three sections): fix structure, and capability should first grow as a power law with scale, then flatten or turn negative after crossing some wall; and this shape must be reproduced in at least three mutually unrelated domains before it deserves to be called a cross-domain law. How to kill it: in strictly controlled comparisons, scale extends linearly along the same slope by more than tenfold without peaking or bending — the three sections degrade to the weak claim of "local power law."

Prediction two (structural jump under equal parameters): under identical parameters, compute, and data, changing only the topology from flat to hierarchical/routed should produce a discontinuous step in capability rather than a smooth gradient. How to kill it: in equal-budget comparisons, the end-to-end performance of the two structures falls in the same confidence interval and is stably reproduced across multiple seeds — that would mean structure affects only an efficiency constant, and this paper exaggerated it.

Prediction three (mismatch must be negative + cross-domain collapse): adding scale under a mismatched structure should stably worsen (after multi-seed recomputation "8 slower than 1" still holds); further, after normalizing the scale axes of each domain by structural characteristic quantities, the seven curves should collapse onto the same family of three-section curves. How to kill it: after ≥ 30 -seed recomputation the "the more added the slower" difference includes zero within the 95% confidence interval; or the seven normalized curves cannot collapse together at all — then this paper honestly retreats to "multi-domain analogy," and "everything has" is downgraded to "each has its own law." This is the most aggressive and most kill-worthy claim of the paper.

6 What This Means for Building Systems

First, settle structure before discussing scale. The order of architectural decisions should be reversed: not "first decide how large a model, then casually build a structure," but first choose the right topology, routing, and module boundaries, then choose scale on the curve prescribed by that structure. Trying to run farther on the wrong curve only hits negative returns faster.

Second, ask three questions before "adding people and machines." Will communication surge quadratically with headcount (USL's κ)? Does the new member bring new information, or a replica of the strongest member (evidence three: replicas produce no value)? Has this scale hit an energy, latency, or trust wall?

Third, treat the turning point as a friend, not a failure. The brain uses cortical folding for the wiring wall, the chip uses multiple cores for the power wall, and the model uses MoE routing for the parameter wall — every saturation is a prompt that it is time to change structure. The scalability of a system is, in essence, its ability to "actively switch structure."

Fourth, remain restrained about inferences toward AGI. If this paper holds, general intelligence will not be the result of "the same network growing infinitely large," but more likely "a new structure + choosing the right scale on its curve." We do not predict when AGI arrives, but only offer a stance: **treating "finding the right structure" itself as an engineering problem that can be searched and measured is more worth investing in than piling scale on a fixed structure.**

7 Where We Might Be Wrong

The lifespan of a flagship paper depends on whether it writes the harshest objections up front.

1. **The turning point may be a measurement illusion.** Schaeffer et al. have shown that large-model "emergence" largely comes from the choice of non-continuous metrics. Our defense is: the six-orders-of-magnitude difference in topology fan-in is deterministic mathematics (closed-form formulas consistent with explicit simulation), not subjective scoring; but the throughput hump was indeed measured at only four discrete levels and must be recomputed with denser scales. When a

"phase transition" cannot be distinguished from a "measurement artifact," it does not deserve to be called a phase transition.

2. **Some walls are physical and cannot be bypassed by changing structure.** The end of Dennard scaling is a physical result of leakage and threshold voltage, and hierarchical topology can only push the communication wall back, not eliminate it. This paper does not claim structure is omnipotent.
 3. **Structure itself is also learned.** The lottery-ticket hypothesis (Frankle & Carbin, 2019) points out that training screens out sparse "winning" substructures from randomly initialized networks; biological neural connections are also shaped by learning. In our comparisons, structure was manually arranged before the experiment began, so they only prove that "**designed** structure affects the scale curve," not that structure "**emerging** during training follows the same law." Filling this gap requires continuously tracking structural metrics such as modularity and path length during training — a clear next step.
 4. **Analogies have boundaries.** Metabolic $3/4$ is the result of three billion years of evolution and physical transport, while agent systems are products of human design; urban scaling laws themselves have data-scope disputes; Metcalfe's law "network value grows quadratically with node count" has been pointed out to overestimate the value of heterogeneous connections. We only borrow the "shape," not the "constants": $3/4$, 0.076, 1.2, and 9 are not the same number and never will be.
 5. **All engine figures are only single-seed pilots on a CPU prototype.** The judge is a rule rather than a real vision model, the physical environment is a point-mass proxy, and the I/O curve is simulated with wait time rather than a real interface. They can prove "the shape of the mechanism holds in a simplified world," not "the constants hold in a real system." This paper therefore positions itself as **a research program plus preliminary silicon evidence**, not a proven theory.
-

8 Conclusion

P0 says: spatial structure is the intelligence. This paper adds the other half it froze: **structure determines the shape of the growth curve, scale determines which section of the curve the system stands on; adding scale on a wrong structure yields negative returns.**

From metabolism's $3/4$ power, to cities' 1.2 and 0.8, to the brain's cortical folding, to the chip's power wall, and to the "2 agents fastest, 8 slowest" hump in our engine, different domains repeatedly exhibit the same three-section curve. But we insist on claiming only "shape isomorphism," not "universal constants" — this far weaker claim is the one that can be taken seriously by experiment.

There is a saying in the field: we are not creating intelligence, we are translating it. In blood vessels, brains, and chemical bonds, nature has already trial-and-erred for billions of years on "how structure and scale work together"; what engineering must do is translate it domain by domain. What this paper offers is not an answer but a set-up target: predictions one, two, and three can kill it at any time, and until killed it deserves serious testing.

References

AI/systems literature reuses this volume's "Master Reference Library" (verified 2026-09-19); the following entries were verified item by item on 2026-09-19/20, and those that cannot be verified are marked [CITATION NEEDED] and are not treated as real citations.

1. West, G. B., Brown, J. H., & Enquist, B. J. (1997). A General Model for the Origin of Allometric Scaling Laws in Biology. *Science*, 276(5309), 122–126. DOI:10.1126/science.276.5309.122.
2. Kleiber, M. (1932). Body size and metabolism. *Hilgardia*, 6, 315–353; and (1961) *The Fire of Life*. Wiley. (Counter: Hulbert et al. point out 3/4 is only an empirical approximation.)
3. Bettencourt, L. M. A. et al. (2007). Growth, innovation, scaling, and the pace of life in cities. *PNAS*, 104(17), 7301–7306. DOI:10.1073/pnas.0610172104.
4. Herculano-Houzel, S. (2012). The remarkable, yet not extraordinary, human brain as a scaled-up primate brain. *PNAS*, 109(Suppl 1), 10661–10668. DOI:10.1073/pnas.1201895109.
5. Amdahl, G. M. (1967). Validity of the Single Processor Approach. *AFIPS SJCC*.
6. Gunther, N. J. (2008). A General Theory of Computational Scalability Based on Rational Functions. arXiv:0808.1431 (USL).
7. Dennard, R. H. et al. (1974). Design of ion-implanted MOSFET's with very small physical dimensions. *IEEE JSSC* (its scaling failed around 2005–2007, and the industry shifted to multi-core).
8. Kaplan, J. et al. (2020). Scaling Laws for Neural Language Models. arXiv:2001.08361.
9. Hoffmann, J. et al. (2022). Training Compute-Optimal Large Language Models (Chinchilla). arXiv: 2203.15556.
10. Fedus, E., Zoph, B. & Shazeer, N. (2021). Switch Transformers. arXiv:2101.03961.
11. Liang Wenfeng et al. (2025). Native Sparse Attention. arXiv:2502.11089.
12. Schaeffer, R., Miranda, B. & Koyejo, S. (2023). Are Emergent Abilities of Large Language Models a Mirage? arXiv:2304.15004.
13. Frankle, J. & Carbin, M. (2019). The Lottery Ticket Hypothesis. *ICLR*. arXiv:1803.03635.
14. Briscoe, B., Odlyzko, A. & Tilly, B. (2006). Metcalfe's Law is Wrong. *IEEE Spectrum*, 43(7), 26–31 (an economic analogy, not a physical law).
15. Dunbar, R. (1990s) Dunbar's number is about 150 (the causal direction is disputed; cite with caution).
16. Vaswani, A. et al. (2017). Attention Is All You Need. arXiv:1706.03762.
17. Smith, R. G. (1980). The Contract Net Protocol. *IEEE Trans. Computers*.

[CITATION NEEDED] : a peer-reviewed measured exponent for battery-pack scale; the exact volume/pages of the original Dennard 1974 and Dunbar papers.

Appendix: Provenance Table for the UDOS Figures in This Paper

Figure	Data file	Grade
Fan-in 20,971,520→9, rounds 8,388,608→14 (six levels)	matrix_scale_demo.json	cpu-proto (first two measured)
CPU throughput 153.208/244.172/158.386/106.485; I/O 19.94/39.88/79.69/159.26 near-linear	agent_scaling.json	cpu-proto (single seed)
Quality saturation $a=0.009252, b=-4.233 \times 10^{-5}$, best single 0.009196; equal weight 0.2131	agent_scaling.json	$k \leq 2$ verified, rest cpu-proto
Yield 0.667→0.854; state units 157→215; density 7.80→11.38; same-pool greedy ablation 0.7917/0.889	egodata_flywheel_demo.json	cpu-proto (48 rounds/144 minutes, oracle 226 units)
Width 64/128/256 = 95,348/271,476/967,796 parameters	model_size_convergence.json	verified
Stigmergy 60 vs Contract Net 240; $n \geq 3f+1$	udos7/topology/	cpu-proto
Multi-view single-view IoU 0.7147/coverage 0.7456/precision 0.9472/depth MAE 0.3186; eight-view IoU 0.8897/coverage 0.9957	spatial_novelview_demo.json	cpu-proto (8 training views/6 held out/ voxel 0.1 m)

(End of full text. P0b is the second axis "scale" of the flagship trilogy; the structure axis is in P0, and the self-boundary axis is in P0c.)